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R2 Data Catalog >

Connect to Iceberg engines >

Spark (PySpark)

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Spark (PySpark)

Below is an example of using [PySpark ↗](#) to connect to R2 Data Catalog.

Prerequisites

- Sign up for a [Cloudflare account ↗](#).
- [Create an R2 bucket](#) and [enable the data catalog](#).
- [Create an R2 API token](#) with both [R2 and data catalog permissions](#).
- Install the [PySpark ↗](#) library.

Example usage

Python

```
from pyspark.sql import SparkSession

# Define catalog connection details (replace variables)
WAREHOUSE = "<WAREHOUSE>"
TOKEN = "<TOKEN>"
CATALOG_URI = "<CATALOG_URI>"

# Build Spark session with Iceberg configurations
spark = SparkSession.builder \
    .appName("R2DataCatalogExample") \
    .config('spark.jars.packages', 'org.apache.iceberg:iceberg-spark-runtime-3.5_2.12:1.6.1,org.apache.iceberg:iceberg-spark-runtime-3.5_2.12:1.6.1') \
    .config("spark.sql.extensions",
"org.apache.iceberg.spark.extensions.IcebergSparkExtensions") \
    .config("spark.sql.catalog.my_catalog",
"org.apache.iceberg.spark.SparkCatalog") \
    .config("spark.sql.catalog.my_catalog.type", "r2") \
    .config("spark.sql.catalog.my_catalog.uri", CATALOG_URI) \
    .config("spark.sql.catalog.my_catalog.warehouse", WAREHOUSE) \
    .config("spark.sql.catalog.my_catalog.token", TOKEN) \
    .config("spark.sql.catalog.my_catalog.header.X-Iceberg-Delegation", "vended-credentials") \
```

```
.config("spark.sql.catalog.my_catalog.s3.remote\n>false") \  
.config("spark.sql.defaultCatalog", "my_catalog\n.getOrCreate()\nspark.sql("USE my_catalog")
```

Create namespace if it does not exist

```
spark.sql("CREATE NAMESPACE IF NOT EXISTS default
```

Create a table in the namespace using Iceberg

```
spark.sql("""\n    CREATE TABLE IF NOT EXISTS default.my_table (\n        id BIGINT,\n        name STRING\n    )\n    USING iceberg\n""")
```

Create a simple DataFrame

```
df = spark.createDataFrame(\n    [(1, "Alice"), (2, "Bob"), (3, "Charlie")],\n    ["id", "name"]\n)
```

Write the DataFrame to the Iceberg table

```
df.write \  
    .format("iceberg") \  
    .mode("append") \  
    .save("default.my_table")  
  
# Read the data back from the Iceberg table  
result_df = spark.read \  
    .format("iceberg") \  
    .load("default.my_table")  
  
result_df.show()
```

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